Project Title
Mini Search Engine / Information-Retrieval System

Project Report submitted for the partial fulfilment of grade for the course

Data Structures and Algorithms -3 (25CS2103E)

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ABSTRACT

This project presents the design and implementation of a custom mini search engine from first principles, focusing on core information-retrieval algorithms without relying on external search libraries. The system efficiently indexes a document corpus to support four primary query classes: exact keyword matching, phrase and substring retrieval, fuzzy string matching, and document-similarity ranking. To achieve this, the architecture utilizes hand-implemented data structures, including an inverted index for fast keyword lookups, suffix arrays for substring queries, dynamic programming for Levenshtein distance in fuzzy searches, and TF-IDF vectorization with cosine similarity for relevance ranking. The documentation thoroughly details these algorithmic choices, providing rigorous justifications for their asymptotic time and space complexities across the indexing, querying, and scaling phases. Ultimately, this work delivers a comprehensive framework that illuminates the mechanical design, algorithmic foundations, and performance trade-offs inherent in modern search systems.

Signature of the Guide